SpcTopics Assignment 2: Glucose Control Part 2

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1 Introduction

The previous assignment introduced a simplified mathematical model of glucose regulation, which describes how glucose levels respond to external input through a feedback control mechanism. By converting the original integral-differential model into a two-dimensional system of ordinary differential equations (ODEs) and applying nondimensionalization, then a reduced model was obtained that allowed for numerical investigation. Equilibrium solutions were calculated, and their stability was examined using Jacobian analysis and eigenvalues. Furthermore, numerical simulations demonstrated that the system's behavior depends on both its parameters and initial conditions, with some trajectories approaching stable equilibrium while others moved away from them. Building on this analysis, the present assignment extends the investigation of the glucose control model by examining one-parameter bifurcation diagrams and exploring the conditions under which the model is controllable.

2 One-Parameter Diagrams

A one-parameter diagram illustrates how the equilibrium solutions of a dynamical system change as one parameter is varied while all other parameters are held constant. A python script was constructed studying the 2-D dynamical system with the variables u and v and parameters a,b,c, and G. It computed the systems two equilibrium branches analytically, then plotted them against G for $a = 0.1$ and $a = 0.5$.

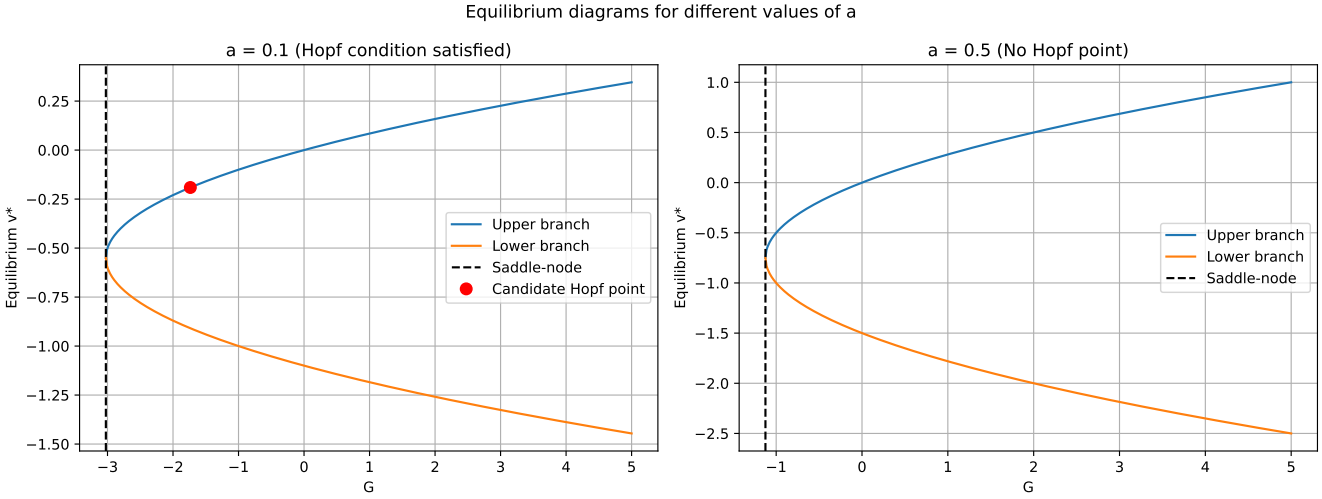
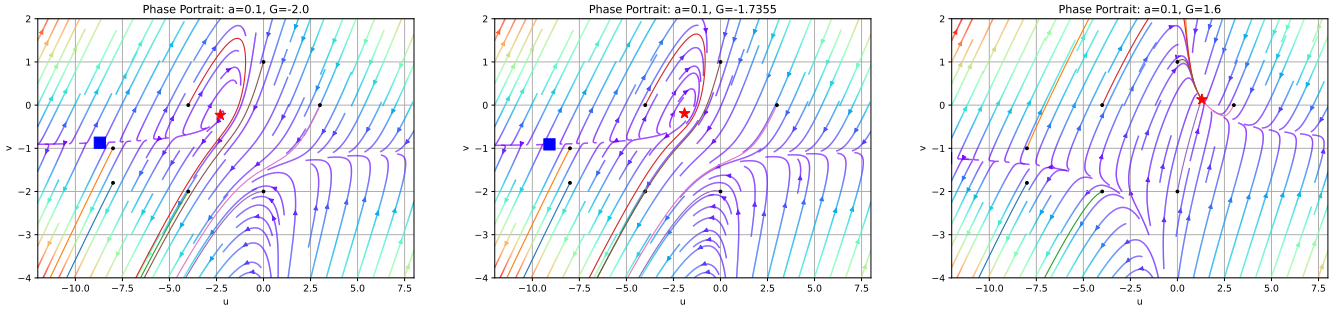


Figure 1: One-parameter diagram of G that demonstrates the appearance of a Hopf Bifurcation.

Figure 1 demonstrates how the equilibrium branches change as the parameter G varies for two different values of a. For $a = 0.1$, a Hopf bifurcation occurs along the upper equilibrium branch, indicating a change in the stability of the equilibrium and the potential emergence of periodic oscillations, existing when $a < \sqrt{2} - 1 \approx 0.414$, which is why $a = 0.1$ has one and $a = 0.5$ doesn't. Expanding on that information, when $a = 0.5$, no Hopf bifurcation is observed. This suggests that changing the parameter a influences whether the system undergoes a Hopf bifurcation as G varies.

3 Making the Model Controllable

Controllability describes the ability to influence a dynamical system through an external input and move the system from an initial state to a desired target state within a finite amount of time. In the context of the glucose control model, this involves investigating how the parameters and initial conditions influence the system's ability to reach a desired state. The stability of the equilibrium points and the behavior of the trajectories provide useful information when examining this question.



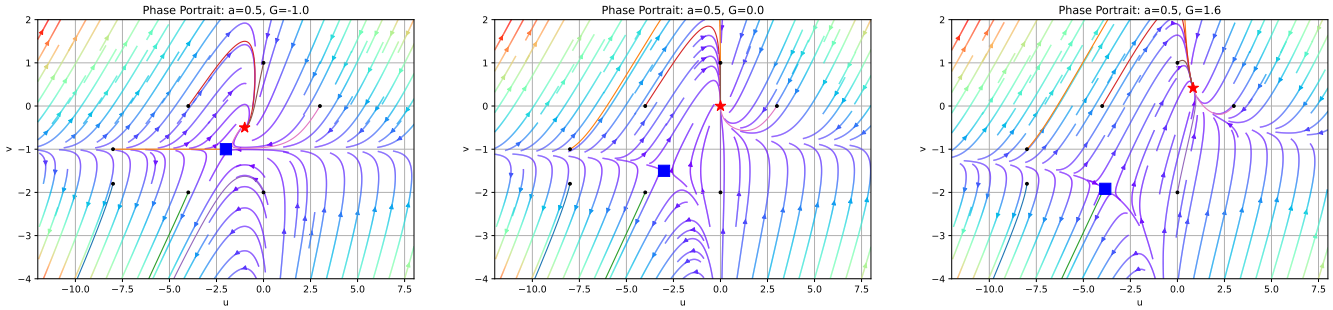
(a) $G = -2.0$ (b) $G = -1.7355$ (c) $G = 1.6$

Figure 2: Phase portraits for $a = 0.1$ at three values of G.

For $a = 0.1$, changing the value of G produces noticeable changes in the stability and behavior of the system. At $G = -2.0$, the phase portrait demonstrates an unstable spiral, where trajectories spiral towards the equilibrium point. As G approaches the Hopf bifurcation value of approximately -1.7355, the stability of the equilibrium changes. At the bifurcation point, the real part of the complex eigenvalues reaches zero. For $G = 1.6$, the phase portrait instead demonstrates a stable node, where trajectories approach the equilibrium without spiraling.

Table 1: Equilibrium stability for $a = 0.1$ at different values of G.

G	Equilibrium	Eigenvalues	Classification
-2.0	0	$0.2141 \pm 0.7710i$	Unstable spiral
-2.0	1	$-0.0847,\ 7.5565$	Saddle
-1.7355	0	$-0.00003 \pm 0.8475i$	Near Hopf
-1.7355	1	$-0.0899,\ 7.9899$	Saddle
1.6	0	$-0.4401,\ -3.0907$	Stable node
1.6	1	$-0.1178,\ 11.5486$	Saddle



(a) $G = -1.0$ (b) $G = 0.0$ (c) $G = 1.6$

Figure 3: Phase portraits for $a = 0.5$ at three values of G.

For $a = 0.5$, the system does not exhibit a Hopf bifurcation as G varies. At $G = -1.0$, equilibrium 0 is a stable spiral, while equilibrium 1 is a saddle, where trajectories approach along one direction and move away along another. As G increases to 0 and then to 1.6, the locations of the equilibrium points change. However, unlike the case where

$a = 0.1$, no Hopf bifurcation occurs. Therefore, the equilibrium points are to be classified individually according to their stability, using the eigenvalues of the Jacobian matrix.

Table 2: Equilibrium stability for $a = 0.5$ at different values of G.

G	Equilibrium	Eigenvalues	Classification
-1.0	0	$-0.5 \pm 0.5i$	Stable spiral
-1.0	1	$-0.5,\ 1.0$	Saddle
0.0	0	$-1.0,\ -1.5$	Stable node
0.0	1	$-0.5811,\ 2.5811$	Saddle
1.6	0	$-2.9642,\ -0.7876$	Stable node
1.6	1	$-0.6053,\ 3.8570$	Saddle

3.1 Controllability

A system is controllable if an outside input can be used to move it from any starting state to any target state. In this model, the input changed is G. Before, G has been kept constant, but here it will changed with time. The system of the Glucose control is

$$\dot{u} = -au + bv + c\, f(u, v),$$
$$\dot{v} = f(u, v), \qquad f = -v - u(v + 1) + G(t).$$

Near an equilibrium (u^*, v^*), the system behaves like a linear system:

$$\dot{x} = Ax + B\, \delta G,$$

where $x = (u, v)^T$, A is the Jacobian matrix at the equilibrium, and

$$A = \begin{pmatrix} -a - c(v^* + 1) & b - c(u^* + 1) \\ -(v^* + 1) & -(u^* + 1) \end{pmatrix}, \qquad B = \begin{pmatrix} c \\ 1 \end{pmatrix}.$$

Here B tells how G acts on u and v.

To test controllability, a matrix, $\mathcal{C} = [\, B \;\; AB \,]$, is constructed. If its determinant is not zero, the input can reach both u and v, and the system is controllable. Calculating the determinant gives

$$\det \mathcal{C} = ac - b.$$

This answer does not depend on the equilibrium point or on G. So, the system is controllable whenever $b \neq ac$. This holds for any starting state and for whether or not a Hopf bifurcation occurs. In the original parameters, $b - ac = \delta\lambda\bar{v}\,\beta/\gamma^2$, so the condition just means $\beta \neq 0$: the part of the model that remembers past glucose levels must be present. With $b = c = 1$, the condition becomes $a \neq 1$, so both $a = 0.1$ and $a = 0.5$ are controllable.

4 Conclusion

Two one-parameter diagrams were compared for this assignment. For $a = 0.1$, a Hopf bifurcation appears on the upper branch at $G \approx -1.7355$. For $a = 0.5$ there is none, because a Hopf point only exists when $a < \sqrt{2} - 1$. When G is held constant, whether the system settles at a stable equilibrium depends on both G and the starting point. If G is allowed to change with time, the model is controllable whenever $b \neq ac$ (equivalently $\beta \neq 0$). Both $a = 0.1$ and $a = 0.5$ meet this condition, regardless of the starting point.